NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 09-04-26**
Time: 3 Hrs **Max. Marks: 180**

09-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-4(P1)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 14 – 17)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 18 – 21)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 22 – 24)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 25 – 30)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 31 – 34)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 35 – 38)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 39 – 41)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 42 – 47)	Questions with **Non-Negative Integer** type	+4	0	6	24
Sec – IV (Q.N : 48 –51)	Questions with **MATRIX MATCH**	+3	-1	4	12
Total				**17**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. A person has 6 cards A,K,2,Q,10,9. The person randomly draws the cards one by one with replacement till he gets 3 consecutive A. If P_n represents the probability that atleast n cards are drawn, then

A) $216P_5 - 180P_4 = 5P_2 + 30P_3$ B) $180P_5 - 216P_4 = 30P_2 + 5P_3 1$

C) $216P_5 + 180P_4 = 5P_2 - 30P_3 2$ D) $180P_5 + 216P_4 = 30P_2 - 5P_3$

2. Let $x = \sqrt[3]{\frac{4}{25}}$. There is a unique value of y such that $0 < y < x$ and $x^x = y^y$. The value of y, expressed in the form $\sqrt[c]{\frac{a}{b}}$, where a and b are relatively prime positive integers and c is a prime number then $a + b - c =$

A) 1534 B) 3154 C) 5413 D) 4513

3. Consider two points $A(x_1, y_1)$ and $B(x_2, y_2)$ on the graph of the function $y = \frac{1}{x}$ such that $0 < x_1 < x_2$ and $|AB| = 2.|OA|$ (O is the origin, i.e., $O(0,0)$. Let C be the midpoint of the segment AB.

A) the angle between the x – axis and the ray OA is equal to three times the angle between x – axis and the ray OC.
B) the angle between the x – axis and the ray OA is equal to two times the angle between x – axis and the ray OC.
C) the angle between the x – axis and the ray OA is equal to the angle between x – axis and the ray OC.
D) the angle between the x – axis and the ray OA is equal to Four times the angle between x – axis and the ray OC.

4. Ten women sit in 10 seats in a line. All of the 10 get up and then reseat themselves using all 10 seats, each sitting in the seat she was in before or a seat next to the one she occupied before. The number of ways can the women be reseated is

A) 89 B) 90 C) 2^{10} D) 91

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SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. Tangents are drawn from $\left(-\frac{\pi}{2},0\right)$ to $f:[0,\infty)\to R; f(x)=\sin x$ and $(a_i,\sin a_i)$ are the points of contact of these tangents for $i=1,2,3,....$ and $a_{i+1}>a_i$. Choose the correct option(s)

A) $a_5+\frac{\pi}{2}=\tan a_5$ B) $a_{10}-a_9<\pi$ C) $a_{11}-a_9>2\pi$ D) $\sin a_{10}-\sin a_9<0$

6. Curves A,B,C and D are defined in the plane as follows:

$A=\left\{(x,y);x^2-y^2=\frac{x}{x^2+y^2}\right\}; B=\left\{(x,y);2xy+\frac{y}{x^2+y^2}=3\right\};$

$C=\left\{(x,y);x^3-3xy^2+3y=1\right\}; D=\left\{(x,y);3x^2y-3x-y^3=0\right\}$, then

A) $(A\cap B)=(C\cap D)$ B) $(A\cap B)\neq(C\cap D)$

C) $\frac{n(A\cap B)}{n(C\cap D)}=2$ D) $\frac{n(A\cap B)}{n(C\cap D)}=4$

7. Let A and B be real $n\times n$ matrices such that $A^2+B^2=AB$. If $BA-AB$ is an invertible matrix then possible value (s) of n is/are

A) 30 B) 31 C) 32 D) 33

SECTION-III (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. Functions f and g are quadratic, $g(x)=-f(100-x)$ and the graph of g contains the vertex of the graph of f. The four x-intercepts on the two graphs have $x-$coordinates $x_1,x_2,x_3,$ and x_4, in increasing order, and $x_3-x_2=150$. The value of x_4-x_1 is $m+n\sqrt{p}$, where m,n ,and p are positive integers, and p is not divisible by the square of any prime. Then $m-n+p=$

For More Join: @IITJEE_Advanced

9. A line touches the circles $x^2+y^2=81$ & $x^2+y^2-16x-16y+112=0$ at A&B respectively. P is one of the points common to both circles& is farthest from AB. The circumradius of ΔPAB is

10. Consider a parabola $P: x^2=4ky$ and a circle $C: x^2+y^2-2\alpha x+6y+9=0$ (where $\alpha>0$). The line of shortest distance between them cuts the parabola P at the end point of latus rectum in first quadrant, and if $k=1$ then $\alpha=$__

11. $y=x$ is tangent to an ellipse having foci at $(2,3)$ and $(5,7)$ then Length of semi –minor axis of the given ellipse is

12. Consider a fair coin and a fair 6-sided die. The die begins with the number 1 face up. A step starts with a toss of the coin: if the coin comes out heads, we roll the die; otherwise (if the coin comes out tails), we do nothing else in this step. After 5 such steps, the probability that the number 1 is face up on the die, expressed as a fraction in reduced form is $\frac{\alpha}{\beta}$ then $\alpha+\beta=$

13. The complex numbers ω and z satisfy the equations $|\omega|=5, |z|=13$, and $52\omega-20z=3(4+7i)$. The value of the product $\omega z=a+ib$ where $(a,b\in R)$ then $|a+b|=$

SECTION-IV (Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

14. Let $\sum_{r=1}^{\infty}\frac{1}{r^2}=a$ and $f(x)=\frac{\ln x}{1-x}, g(x)=\left(\frac{x\ln x}{1-x^2}\right)$. Let $I_1=\int_0^1 f(x)dx$ and $I_2=\int_0^1 g(x)dx$,

Match each entry in List-I to the correct entry in List-II.

List-I		List-II	
P)	I_1	1)	$-a$
Q)	I_2	2)	$-\frac{3a}{4}$
R)	I_1-I_2	3)	$-\frac{a}{4}$
S)	$\lim_{n\to\infty}\sum_{r=1}^{n}\left(\frac{r(\ln r-\ln n)}{n^2-r^2}\right)=$	4)	$+a$
		5)	$\frac{-a}{2}$

For More Join: @IITJEE_Advanced

A) $(P)\to(4),(Q)\to(2),(R)\to(5),(S)\to(1)$

B) $(P)\to(5),(Q)\to(4),(R)\to(3),(S)\to(1)$

C) $(P)\to(1),(Q)\to(3),(R)\to(2),(S)\to(3)$

D) $(P)\to(1),(Q)\to(2),(R)\to(2),(S)\to(1)$

15. For non-negative integers a and b, let $I(a,b)=\int_0^{\frac{\pi}{2}}\cos^a x\cos bx\,dx$.

Column-I		Column-II	
P)	$I(0,5)=$	1)	$\frac{1}{5}$
Q)	$I(1,4)=$	2)	$\frac{1}{5}I(2,1)$
R)	$I(2,4)=$	3)	$\frac{1}{4}\left[I(1,3)-I(1,5)\right]$
S)	$I(3,2)=$	4)	$\frac{1}{5}I(0,3)$
		5)	$\frac{1}{3}I(2,1)$

A) $(P)\to(1),(Q)\to(4),(R)\to(3),(S)\to(2)$

B) $(P)\to(2),(Q)\to(1),(R)\to(5),(S)\to(3)$

C) $(P)\to(3),(Q)\to(1),(R)\to(5),(S)\to(2)$

D) $(P)\to(1),(Q)\to(4),(R)\to(1),(S)\to(3)$

16. The function f is defined, for $x>1$, by $f(x)=\int_1^x\sqrt{\frac{t-1}{t+1}}dt$.

Match each entry in List-I to the correct entry in List-II.

List-I		List-II	
P)	for $x>2$, $\int_2^x\sqrt{\frac{u-2}{u+2}}du=$	1)	$2f\left(\frac{1}{2}x+1\right)$.
Q)	for $x>0$, $\int_0^x\sqrt{\frac{u}{u+4}}du$	2)	$2f\left(\frac{1}{2}x\right)$.
R)	for $x>5$, $\int_5^x\sqrt{\frac{u-5}{u+1}}du$	3)	$f(3)-f(1.5)$
S)	$\int_1^2\frac{u^2}{\sqrt{u^2+4}}du$	4)	$f(2)-f(1.5)$
		5)	$3f\left(\frac{x-2}{3}\right)$.

For More Join: @IITJEE_Advanced

A) $(P)\to(2),(Q)\to(1),(R)\to(5),(S)\to(3)$

B) $(P)\to(1),(Q)\to(2),(R)\to(5),(S)\to(3)$

C) $(P)\to(1),(Q)\to(2),(R)\to(5),(S)\to(4)$

D) $(P)\to(2),(Q)\to(2),(R)\to(1),(S)\to(4)\ln$

17. Match the following.

Column-I		Column-II	
P)	The number of points of non-derivability of the function $f(x)=\left[\frac{2x}{\pi}\right]\operatorname{sgn}\left(\frac{1}{\{x\}}\right)$ in $(-2,2)$ is (where $[y],\{y\}$ and sgn(y) denote largest integer $\le y$ fractional part of y and signum function of y respectively	1)	7
Q)	If $f:[0,\infty)\to R$ be a function defined as $f(x)=\begin{cases}\left(2\sin\sqrt{x}+\sqrt{x}\sin\left(\frac{1}{x}\right)\right)^x, & x>0\\ K, & x=0\end{cases}$ is continuous at $x=0$ then $7K$ is	2)	3
R)	Let ABC be a variable triangle such that A is $(1,2)$, B and C lie on the line $y=x+\lambda$ (where λ is a real variable except 1). The locus of the orthocentre of triangle ABC is a straight line whose y-intercept is equal to	3)	4
S)	The number of integral values of x satisfying $\frac{\left(2^{x^2}-4\right)(x-4)(x-9)}{x(x-1)}<0$, is	4)	5

A) P-4,Q-1,R-2,S-3

B) P-1,Q-2,R-4,S-3

C) P-4,Q-3,R-2,S-1

D) P-3,Q-2,R-1, S-1

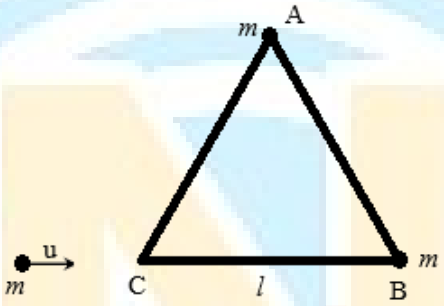
For More Join: @IITJEE_Advanced

PHYSICS **MAX.MARKS: 60**

SECTION–II (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
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Negative Marks: -1 In all other cases.

18. An equilateral massless triangular frame ABC of side length l with two identical point masses each of mass m attached to it at corners A and B is place at rest on a smooth horizontal plane shown in the figure below (top view). A third identical point mass C moving in the same horizontal plane with a speed u along the line CB collides with corner C and sticks to it at t = 0. Which of the following is false in the subsequent motion of the triangular frame?



A) maximum speed attained by the particle at A is $\frac{u}{2}$

B) speed of particle at B at time $t=\frac{\pi l}{\sqrt{3}u}$ is $\frac{\sqrt{5}u}{6}$

C) minimum speed attained by the particle at B is $\frac{u}{6}$

D) particle at C attains its minimum speed at $t=\frac{2\pi l}{\sqrt{3}u}$

19. Current is flowing in a cylindrical wire of radius a along the axis such that the current density varies with distance r perpendicular to axis as $j=\frac{j_o r^2}{a}$, where j_o is a constant. Which of the following statements is false?

A) Current in $r = 0$ to $r=\frac{a}{\sqrt[4]{2}}$ of the cross-section is equal to current in the remaining cross-section.

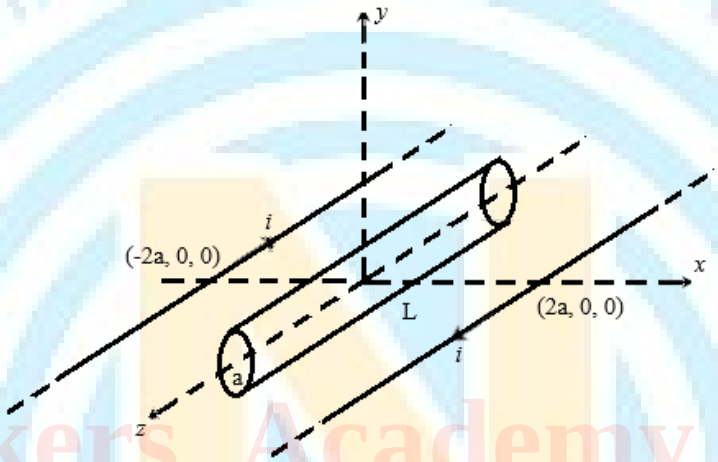
B) The magnitude of magnetic field at the perpendicular distance of $2a$ from the axis is $\frac{\mu_o j_o a^2}{8a}$

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C) At the perpendicular distances of $3a$ and $\frac{a}{\sqrt[3]{3}}$ from the axis the magnitude of magnetic field is same

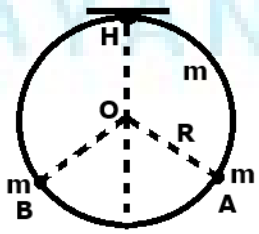
D) The magnitude of magnetic field at the perpendicular distance of $r=\frac{a}{2}$ from the axis is $\frac{\mu_o j_o a^2}{8a}$

20. Two long current carrying straight wires with each carrying a current of i are placed antiparallelly long the lines parallel to z-axis passing through $(-2a,0,0)$ and $(2a,0,0)$ respectively as shown in the figure. If a cylindrical surface of radius a and length L is placed with its axis along z-axis. Find the value of the magnetic flux through the upper half curved surface area of this cylinder, that is, the area that lies above the xz plane.



A) Zero B) $\frac{\mu_o iL}{\pi}\ln 2$ C) $\frac{\mu_o iL}{\pi}\ln 3$ D) $\frac{\mu_o iL}{2\pi}\ln 3$

21. A uniform ring of radius R and mass m with two identical point masses A and B each of same mass m attached to it hinged at a smooth point H and free to rotate in the vertical plane. Points A, B and H form an equilateral triangle. The ring is disturbed by a negligible angle from the stable equilibrium orientation as shown in the figure. Find the time period of small oscillations about the horizontal axis passing through H perpendicular to the plane of the paper.



A) $2\pi\sqrt{\frac{R}{g}}$ B) $2\pi\sqrt{\frac{2R}{g}}$ C) $4\pi\sqrt{\frac{R}{g}}$ D) $4\pi\sqrt{\frac{2R}{g}}$

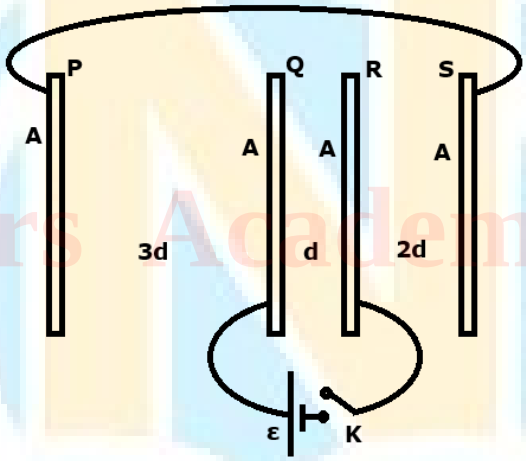
For More Join: @IITJEE_Advanced

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

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Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

22. Four large thin neutral conducting plates P, Q, R and S each of plate area A are placed parallel to each other in vacuum with gaps $3d$, d, $2d$ (where d very small compare to the dimensions of the plates). P and S are connected by a thin conducting wire, whereas an ideal battery of emf ε with an open key K is connected across Q and R as shown in the figure. Now the key is closed, then the charge on



A) P is $-\frac{\varepsilon_o A}{5d}\varepsilon$ B) Q is $\frac{6\varepsilon_o A}{5d}\varepsilon$ C) R is $-\frac{\varepsilon_o A}{d}\varepsilon$ D) S is $-\frac{\varepsilon_o A}{5d}\varepsilon$

23. A calorimeter with an empty inner chamber is initial at the room temperature of 20°C, to determine its water equivalence of this calorimeter 300g of water at 50°C is poured into the outer chamber of the calorimeter and fully sealed. After a long time, the thermometer connected to the water reached a saturation temperature of 30°C. Then, the seal is opened, water is poured out and the calorimeter is allowed to cool down to the room temperature. Now, 400g of fresh water at room temperature is poured into the outer chamber and 100g of an unknown metal heated to 120°C is placed in the inner chamber of the calorimeter and sealed. Once again after waiting for a long period of time the reading in the thermometer connected to the water reached a saturation temperature of 40°C. Which of the following is/are incorrect? [Take specific heat of water as 1cal/g-K]

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A) Water equivalence of the calorimeter system is 800g

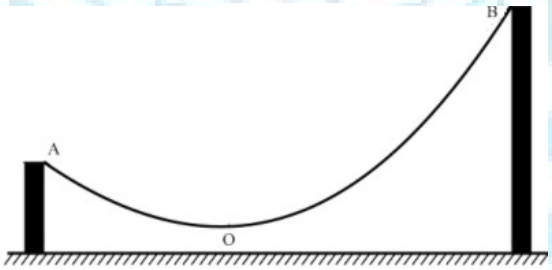
B) Water equivalence of the calorimeter system is 700g

C) Specific heat of the unknown metal is 2.5cal/g-K

D) Specific heat of the unknown metal is 5cal/g-K

24. A thin uniform rope AOB of mass per unit length λ is connected at ends A and B to two vertical fixed walls such that O is the bottom-most point on the rope. The length of the rope from O to B is $3l$ and vertical height gap between O and B is l. Then which of the following is/are correct

[g is magnitude of acceleration due to gravity]



A) Tension in the rope is minimum at A

B) Minimum tension in the rope is $4\lambda lg$

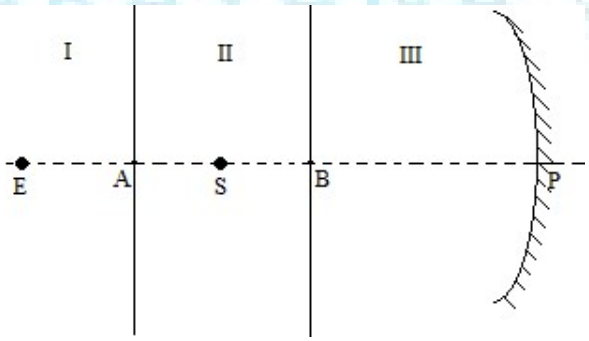
C) Tension in the rope at end B is $5\lambda lg$

D) The shape of the rope is parabolic

SECTION-III (Maximum Marks : 24)

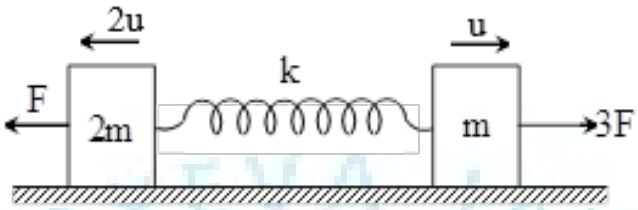
This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

25. Space is divided in to three regions I, II and III by two plane surfaces A and B as shown. Refractive indices of the I, II, III are 4.2, 1.05, and 2.1 respectively. An eye E and source of light S are placed on the principal axis of a concave mirror of focal length 14cm. If EA = 20cm, AS = SB = BP = 7cm. How far from itself does the eye E perceive the image of S in the mirror. (Give the answer in cm)



For More Join: @IITJEE_Advanced

26. Two blocks m and $2m$ are connected by an ideal massless spring of force constant k and placed on a smooth horizontal floor with external constant forces 3F and F acting on m and $2m$ respectively. When the spring is in its natural length the speeds of the blocks are u and $2u$ as shown in the figure. Find the maximum extension in the spring (in cm), if F = 3N, k = 100N/m, u = 1m/s, m = 25grams.



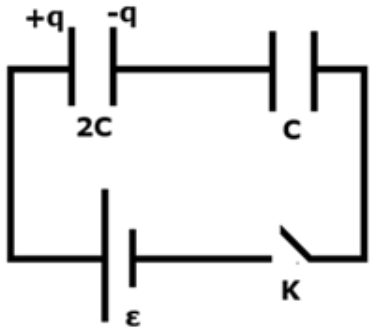
27. The unstable isotope Uranium-233 (atomic number 92) undergoes a sequence of alpha and beta-minus decays to form a stable Lead-205 (atomic number 82). If α, β are the number of alpha and beta-minus decays in this sequence respectively, find the value of αβ.

28. A thin uniform rod of ABC of length 80cm with AB = 20cm and BC = 60cm has a young's modulus of 9GPa and an area of cross-section of 55μm^2. It's placed on smooth horizontal surface and two forces F and 24N are applied on it along the rod in same direction as shown in the figure. At steady state the change in length of the rod AC is zero, then find the value of the force F (in N).



29. A Vernier Caliper with a main scale division length of 0.05mm is designed such that 23 main scale divisions are equal in length to 25 Vernier scale divisions. When a small object's length is measured using this scale, it is found that the largest number on the main scale crossed by the zero of the Vernier scale is 0.60mm and the 21st division of the Vernier scale exactly coincides with one of the main scale divisions. Find the length of the object in microns. [Consider no zero error]

30. Uncharged capacitor C and a pre-charged capacitor 2C with an initial charge $q = \frac{\varepsilon C}{4}$ are connected to an ideal battery of emf ε as shown in the figure with an open key. When the key is closed, the heat generated is $\frac{n^2}{192} C\varepsilon^2$. Find the value of n.

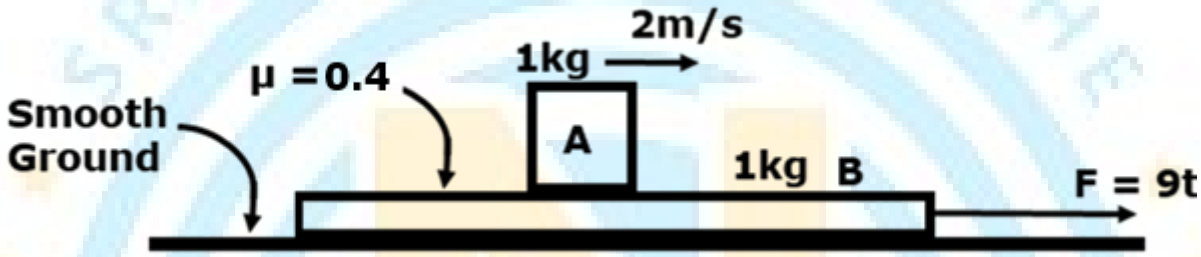


For More Join: @IITJEE_Advanced

SECTION-IV (Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

31. A small block A of mass 1kg initially moving horizontally with a speed of 2m/s is placed on a large plank of B of mass 1kg. The plank initially starts from rest on the smooth ground and is being pulled with a time depend force horizontal force F = 9t as shown in the figure. The coefficient of friction between A and B is 0.4. [Take g = 10m/s^2, assume that A doesn't fall of B and all quantities are measured in SI units]



List-I (Time)		List-II (Speed)	
I)	At $t=\frac{1}{9}s$	P)	Speed of A $=\frac{13}{9}m/s$
II)	At $t=\frac{2}{9}s$	Q)	Speed of A $=\frac{14}{9}m/s$
III)	At $t=\frac{4}{9}s$	R)	Speed of A $=\frac{16}{9}m/s$
IV)	At $t=\frac{10}{9}s$	S)	Speed of B $=\frac{35}{9}m/s$
		T)	Speed of B $=\frac{22}{9}m/s$
		U)	Speed of B $=\frac{10}{9}m/s$

A) I – P, II – U, III – R, IV – S
B) I – Q, II – U, III – R, IV – T
C) I – U, II – T, III – R, IV – S
D) I – Q, II – U, III – P, IV – S

32. An unstable radioactive nucleus A decays into a daughter nucleus B with a decay rate constant λ. The nucleus of B is also unstable and further decays into a stable nucleus C with a decay rate constant 2λ. Initially, the sample contains 2N undecayed nuclei of A, N undecayed nuclei of B and no nuclei of C. Then, match the Lists below

For More Join: @IITJEE_Advanced

List-I (Time)		List-II (Number of Undecayed Nuclei of)	
I)	At $t = \frac{\ln 2}{2\lambda}$	P)	A is $\frac{N}{\sqrt{2}}$
II)	At $t = \frac{\ln 2}{\lambda}$	Q)	A is $\sqrt{2}N$
III)	At $t = \frac{3\ln 2}{2\lambda}$	R)	B is $\frac{7N}{16}$
IV)	At $t = \frac{2\ln 2}{\lambda}$	S)	B is $\frac{3N}{4}$

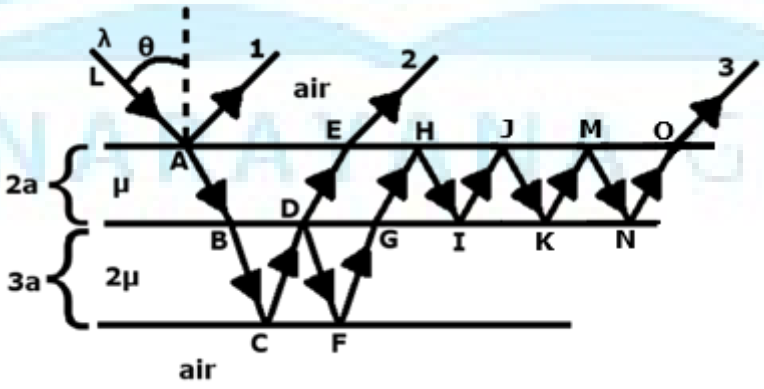
A) I – Q, II – S, III – P, IV – R B) I – Q, II – S, III – R, IV – P

C) I – R, II – S, III – Q, IV – P D) I – R, II – Q, III – P, IV – S

33. Light incident at the boundary of two mediums can undergo partial reflection and partial refraction as its energy splits. A thin film is a perfect example of a medium with two boundaries where light can undergo partial reflection and partial refraction at both the boundaries in a sequential manner. Diagram below shows a light L of wavelength λ in air (refractive index 1) incident at an angle of incidence θ on a stack of two thin films of thickness $2a$ and $3a$ with refractive indices μ (>1) and 2μ respectively placed one above the other.

Light 1 is due to reflection of L at A, Light 2 is the emergent light at E after tracing the path ABCDE and Light 3 is emergent at O after following the path ABCDFGHIJKMNO.

Note that at all these points both phenomena take place but the diagram shows only the phenomenon along the path of interest. For example, at B only refracted light is indicated, even reflected light simultaneously exists and similarly at C only reflected light is shown but not the refracted.



For θ = 0, match the Lists-I which has conditions to List-II that contains the minimum non-zero value of $\frac{\mu a}{\lambda}$:

For More Join: @IITJEE_Advanced

List-I		List-II	
I)	For Light 1 and Light 2 to undergo constructive interference	P)	$\frac{1}{80}$
II)	For Light 1 and Light 2 to undergo destructive interference	Q)	$\frac{1}{40}$
III)	For Light 1 and Light 3 to undergo constructive interference	R)	$\frac{1}{16}$
IV)	For Light 1 and Light 3 to undergo destructive interference	S)	$\frac{1}{32}$

A) I – R, II – S, III – P, IV – Q B) I – S, II – R, III – Q, IV – P

C) I – R, II – S, III – Q, IV – P D) I – S, II – R, III – P, IV – Q

34. A small particle is suspended from a massless string of length R, whose one end is the particle and the other end is fixed. The particle can move in a vertical plane and it's given an initial horizontal speed of $\sqrt{agR}$ when it is at its lowest point. (a is just a number). List-I has different possible values of a and List-II is the ratio of the tension in the string to the weight of the particle at highest point in the particle's journey. Find the correct matching between the lists.

List-I		List-II	
I)	$a = 6.2$	P)	$\frac{2}{5}$
II)	$a = 5.8$	Q)	$\frac{3}{5}$
III)	$a = 1.2$	R)	$\frac{4}{5}$
IV)	$a = 0.8$	S)	$\frac{6}{5}$

A) I – S, II – Q, III – P, IV – R B) I – S, II – R, III – Q, IV – P

C) I – S, II – R, III – P, IV – Q D) I – Q, II – R, III – P, IV – S

For More Join: @IITJEE_Advanced

CHEMISTRY **MAX.MARKS: 60**

SECTION–II (Maximum Marks: 12)

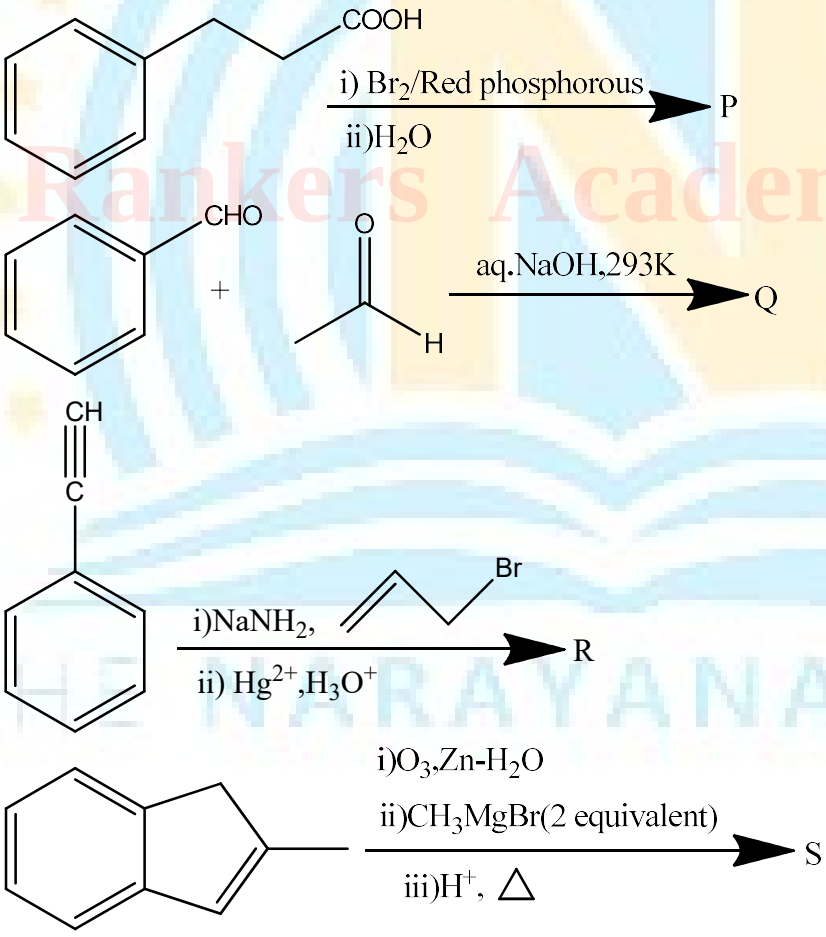
This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

35. Under suitable conditions, nitrobenzene can be reduced electrolytically to N-phenyl hydroxyl amine. In an experiment, a 0.2 g sample containing some nitrobenzene and remaining inert impurity were electrolyzed using a current of 100 mA. It took 6 minutes to reduce the entire nitrobenzene present in the sample. The % purity of sample is (approximately)

Atomic mass: [C-12,N-14,O-16,H-1]

A) 28 B) 5.73 C) 72.4 D) 43.5

36. Monocyclic compound P,Q,R and S are the major products formed in the reaction sequences given below



The product which decolourise bromine water are

A) P only B) P and Q only C) Q,R,S only D) All

For More Join: @IITJEE_Advanced

37. i) $CuSO_4 + KCN(excess) \rightarrow (P)$

ii) $CuCl_2 + KCl(excess) \rightarrow (Q)(mostly)$

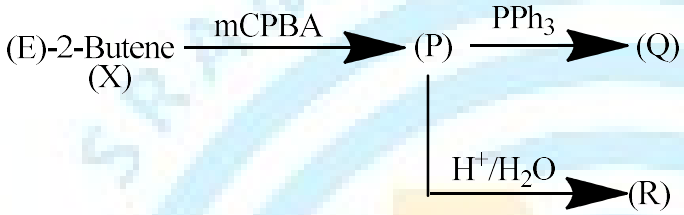
iii) $FeSO_4(aq) + NaNO_2 \xrightarrow{dil.H_2SO_4} (S)$

iv) $MnSO_4 + PbO_2 \xrightarrow[\Delta]{H^+} (V)$

The sum of the spin-only magnetic moments (in B.M) of the complexes formed in the above reactions is

A) 5.6 B) 5.9 C) 3.87 D) 1.73

38.



Which one of the following option is CORRECT?

A) Formation of (P) is stereo selective as well as stereo specific

B) (X) and (Q) are Homomers

C) (R) exists as unresolvable enantiomeric mixture and it is optically inactive

D) $(Q) \xrightarrow[(2)CH_3COOD]{(1)BD_3,THF} (\pm)threo\ product$

SECTION - II (Maximum Marks : 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. 5g sample containing Fe_3O_4 (equimolar mixture of Fe_2O_3 and FeO) and an inert impurity is treated with excess KI in presence of dilute H_2SO_4. Entire iron is converted into Fe^{2+} and I_2 is liberated. Resulting solution is diluted to 200mL. 40mL of diluted solution require 10mL 0.5M $Na_2S_2O_3$ to titrate iodine. Select the correct statement(s)

For More Join: @IITJEE_Advanced

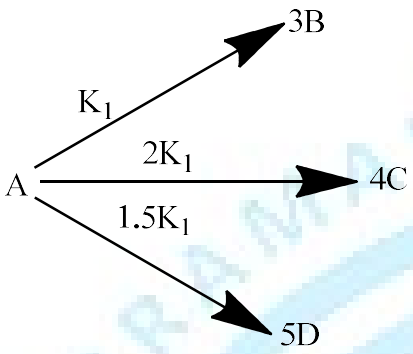
A) Mass % of Fe_2O_3 in sample is 40%

B) Mass % of inert impurities is 42%

C) Mass % of FeO in sample is 28%

D) Mass % of inert impurities is 32%

40.



All reactions are 1st order:

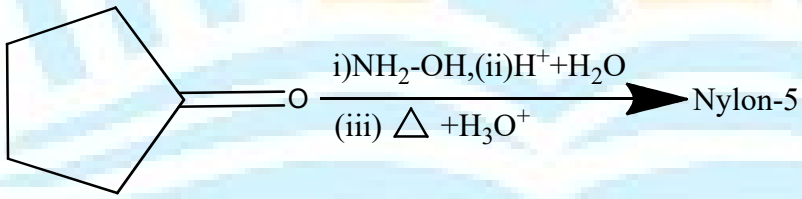
At time $t = t_1 (t > 0), \frac{[B]}{[C]} = \alpha$ and

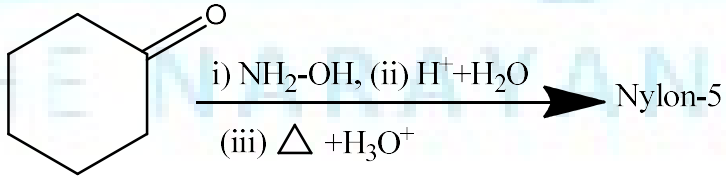
At time $t = t_2 [where\ t_2 \geq t_1], \frac{[C]}{[D]} = \beta$

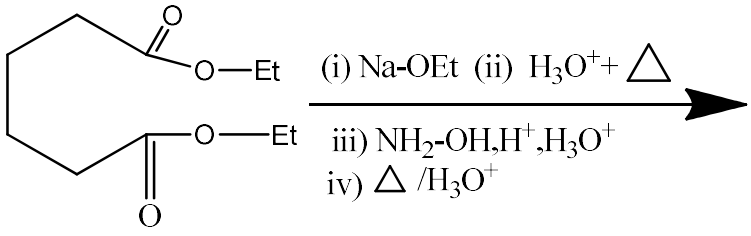
Which of the following is/are correct?

A) $\alpha = 0.5$ B) $\beta = \frac{4}{3}$ C) $\alpha = 0.375$ D) $\beta = \frac{16}{15}$

41. By which of the following reaction sequence can nylon-5 be prepared.

A) 

B) 

C) 

D) all of these

For More Join: @IITJEE_Advanced

SECTION-III (Maximum Marks : 24)

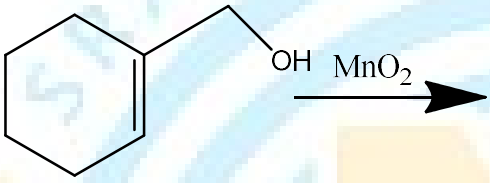
This section contains **SIX (06)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

42. Number of reactions in which aldehyde can be obtained as one of product.

i) $CH_3 - CH_2OH \xrightarrow[300^c\,C]{Cu}$

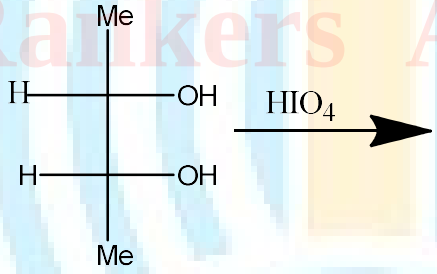
ii) $CH_3 - CH_2 - CH_2 - OH \xrightarrow[Pyridine]{CrO_3}$

iii) $CH_3\text{-}\underset{\underset{O}{\|}}{C}\text{-}Cl \xrightarrow[BaSO_4]{H_2\text{-}Pd}$

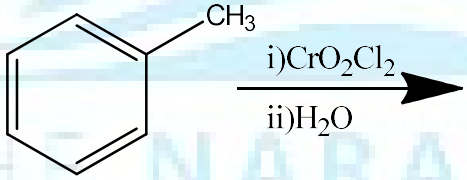
iv) 

v) $CH_3 - C \equiv N \xrightarrow[(ii)\,H_2O]{(i)\,DIBAL-H,\ Low\ temperature}$

vi) $CH_3 - CH = NH \xrightarrow{H_3O^+}$

vii) 

viii) $CH_3 - CH = CH - CH_3 \xrightarrow[Zn]{O_3}$

ix) 

43. For an ideal gas sample (fixed moles taken in fixed volume container) collision frequency (z) and mean free path(λ)

i) Both Z& λ change with change in T

ii) Z changes with change in T

iii) λ Changes with change in T

iv) Both are independent of temperature T

For More Join: @IITJEE_Advanced

v) Heavier gas (more molecular weight) exhibits high Z compared to lighter gas for same molecular diameter.

vi) Heavier gas exhibits more λ compared to lighter gas

vii) Molecular weight of gas does not have any effect of value of λ

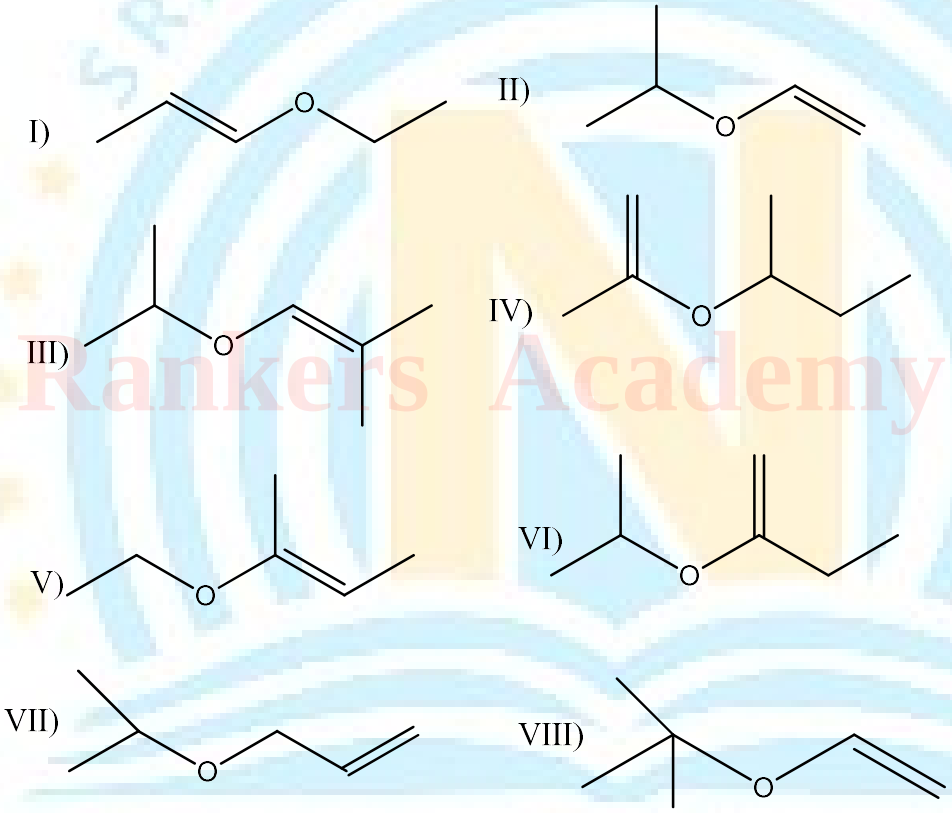
Identify number of correct statement from above?

44. Consider following reactions:

$Na_2S(aq) + Na_2\left[Fe(CN)_5 NO\right] \rightarrow X$ (Purple coloration)

How many linkage isomer are possible for X.

45. In how many of the following ethers both the hydrolyzed products undergo positive iodoform test?



46. 3g of 98% pure sample of $K_XH(C_2O_4)_Y$ is reacted with 100mL of a 0.1 M acidified $KMnO_4$ solution. The excess oxidant required 20mL of 0.5M mohr salt solution for complete reaction. The same mass of another sample of $K_XH(C_2O_4)_Y$ is reacted with 0.4M NaOH. The volume of NaOH required for complete reaction in ml is?

47. Fully charged lead storage battery contains 1.5L of 5M H_2SO_4. If 2.5 amp of current is drawn for 965 minutes, then final left concentration of H_2SO_4 is __M

(Assume volume of battery fluid to be constant)

For More Join: @IITJEE_Advanced

SECTION-IV (Maximum Marks : 12)

This section contains **FOUR (04)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

48. Match the thermodynamic process given List-I with expression given under List-II

	List-I		List-II
P)	Melting of ice at 273K&1 atm	1)	$q=0$
Q)	Expansion of 1 mol of an ideal gas into a vaccum under isolated conditions	2)	$w=0$
R)	Mixing of equal volume of two different ideal gases at constant temperature and pressure in an isolated container	3)	$\Delta S_{sys}>0$
S)	Reversible heating of $H_2(g)$(ideal) at 1atm from 300K to 600K, followed by reversible cooling to 300K at 1atm	4)	$\Delta U=0$
		5)	$\Delta H=0$

A) $P\rightarrow 4;Q\rightarrow 3,4;R\rightarrow 2,3,4;S\rightarrow 2,4$

B) $P\rightarrow 1,4;Q\rightarrow 2,3,4;R\rightarrow 3,4;\ S\rightarrow 1$

C) $P\rightarrow 1,2,3,4;Q\rightarrow 1,2,3;R\rightarrow 3,4;S\rightarrow 1$

D) $P\rightarrow 3;Q\rightarrow 1,2,3,4,5;R\rightarrow 1,2,3,4,5;S\rightarrow 1,2,4,5$

49. Match the column

	List-I		List-II
P)	$\left[Cr(H_2O)_4Br_2\right]^+$	1)	Paramagnetic in nature
Q)	$\left[Cu(H_2NCH_2CH_2NH_2)(CN)_2Cl_2\right]$	2)	Geometrical isomerism is exhibited
R)	$\left[Pt(OX)_2\right]^{2-}$	3)	Central atom in d^2sp^3 hybridization
S)	$\left(Fe(OH)_4\right)^-$	4)	Complex having symmetrical bidentate ligand
		5)	Can show optical isomerism

A) $P\rightarrow 1,2,3;Q\rightarrow 1,2,4,5;R\rightarrow 4;S\rightarrow 1$

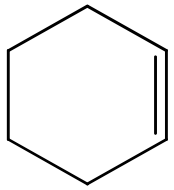
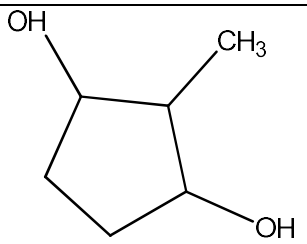
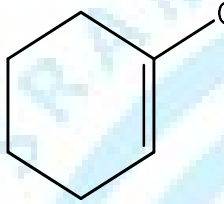
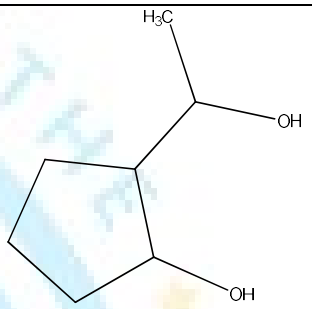
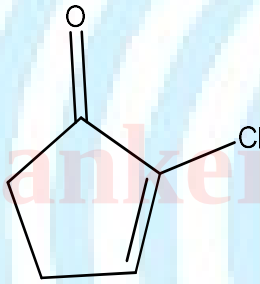
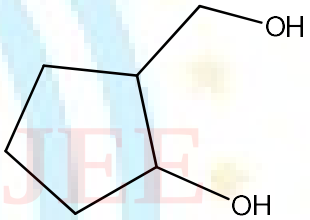
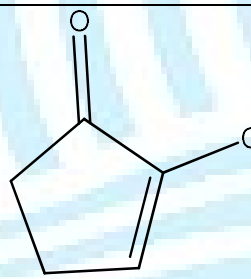
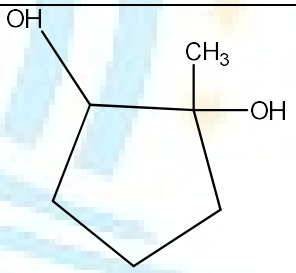
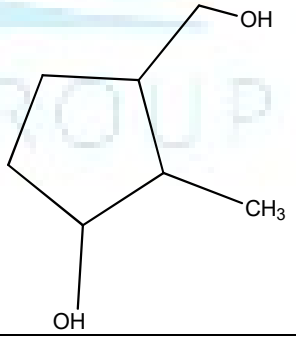
B) $P\rightarrow 1,2,3;Q\rightarrow 1,2,5;R\rightarrow 1,2,3;S\rightarrow 3$

C) $P\rightarrow 2,3,4;Q\rightarrow 1,4;R\rightarrow 4;S\rightarrow 1,5$

D) $P\rightarrow 3,5;Q\rightarrow 4,5;R\rightarrow 1;S\rightarrow 1,2,3$

For More Join: @IITJEE_Advanced

50. **List-I** contains various reaction sequences and **List-II** contains the possible products. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option. (Here PTSA stands for p-Toluenesulfonic acid in List-I)

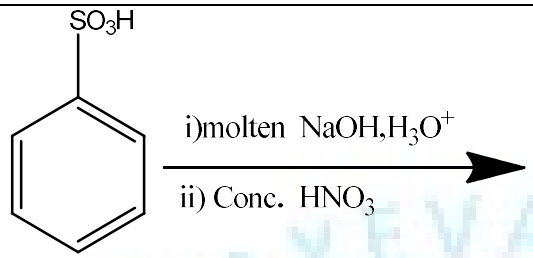
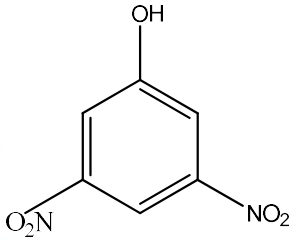
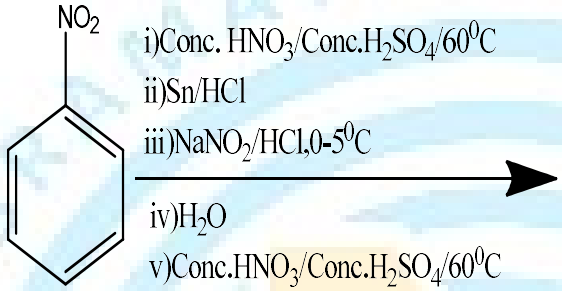
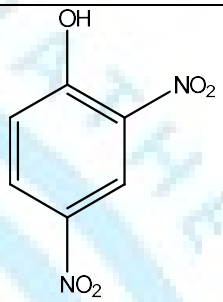
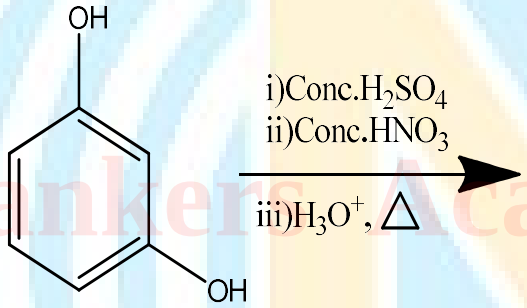
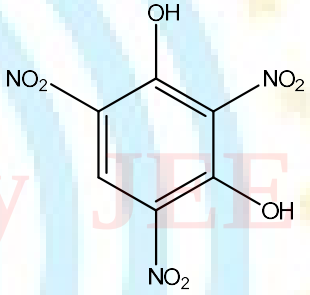
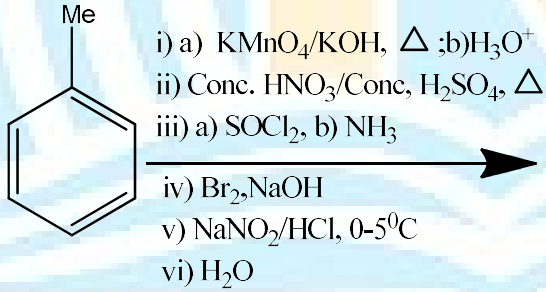
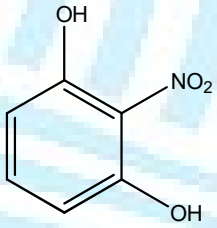
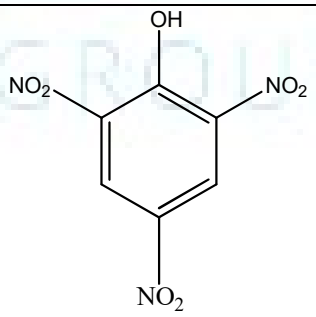
	List-I		**List-II**
P)	 i)O_3,Zn ii)aq.NaOH, Δ iii)ethylene glycol, PTSA → iv) a)BH_3, b)H_2O_2, NaOH v)H_3O^+ vi)$NaBH_4$	1)	
Q)	 i)O_3,Zn ii)aq.NaOH, Δ → iii)ethylene glycol, PTSA iv) a)BH_3, b)H_2O_2, NaOH v)H_3O^+ vi)$NaBH_4$	2)	
R)	 i)ethylene glycol, PTSA → ii) a)$Hg(OAc)_2$,H_2O, b) $NaBH_4$ iii) H_3O^+ iv)$NaBH_4$	3)	
S)	 i)ethylene glycol, PTSA → ii) a)BH_3, b) H_2O_2,NaOH iii) H_3O^+ iv)$NaBH_4$	4)	
		5)	

A) $P \to 3; Q \to 2; R \to 1; S \to 4$ B) $P \to 3; Q \to 5; R \to 4; S \to 1$

C) $P \to 3; Q \to 2; R \to 4; S \to 1$ D) $P \to 2; Q \to 5; R \to 4; S \to 1$

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51. **List-I** contains various reaction sequences and **List-II** contains different phenolic compounds. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

	List-I		List-II
P)		1)	
Q)		2)	
R)		3)	
S)		4)	
		5)	

A) $P \to 2; Q \to 5; R \to 4; S \to 3$ B) $P \to 5; Q \to 3; R \to 4; S \to 1$

C) $P \to 2; Q \to 5; R \to 3; S \to 1$ D) $P \to 5; Q \to 2; R \to 3; S \to 4$

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